DELOITTE FEDERAL FDE RECRUITING · TAKE-HOME ASSIGNMENT #3

FDA CDER Drug Shortage Briefing Tool

FDE Take-Home Assignment — Interviewee Copy

Assignment #3 — FDA / Center for Drug Evaluation and Research (CDER) Scenario

TOOLS	SUBMISSION
Use any tools you have access to, including AI coding assistants	Bring to Interview 2

Background

Dr. Sandra Whitmore is the Deputy Center Director (SES) at FDA's Office of Drug Shortage Staff (ODSS), a division within the Center for Drug Evaluation and Research. Her office monitors the national drug supply, coordinates shortage response with manufacturers, and briefs Congress when shortages become a public health matter.

Next week, she testifies before the Senate HELP Committee on pediatric antibiotic shortages. Her staff needs to enter that hearing with a clear, current, and defensible picture of which drugs are short, why, which manufacturers are implicated, and what the approval landscape looks like.

Until last year, an AI-powered briefing tool was helping her team assemble this information. That tool is no longer in use.

The specific incident that ended it: a staff member asked the tool whether the amoxicillin shortage was resolved. The tool said yes. The answer reached a draft of Dr. Whitmore's Senate testimony. Two days before the hearing, a staff pharmacist noticed the tool had conflated two different status values — one meaning the supply chain had recovered, another meaning the manufacturer had permanently exited the market. A formal correction letter had to be sent to the Senate committee.

A second error — an invented manufacturing site identifier — was caught the night before the hearing by a CDER chemist who recognized the number format was wrong.

The tool was pulled. Staff now cross-reference three separate databases manually, in Excel.

> *The bar is not "can it answer questions." The bar is "can Dr. Whitmore read this aloud in a Senate hearing and stand behind it?"*

You are coming in as an Agent/GenAI Developer and Applied AI Engineer. Dr. Whitmore wants to understand whether modern agentic workflows and LLM systems can help her staff brief senior leadership and Congress more reliably than a manual spreadsheet process. She has asked you to build a lightweight prototype that demonstrates the direction.

The Data

Three CSV files are provided. They represent a synthetic but structurally realistic snapshot of FDA operational data as of early 2025. No real patient or manufacturer data is included.

`drug_shortages.csv` — Current and historical drug shortage records, modeled after FDA's Drug Shortage Database. Each row represents a shortage record with status, reason, and links to the relevant drug application.

`inspections.csv` — Manufacturing facility inspection outcomes, modeled after FDA's Establishment Inspection Report data. Each row represents a completed inspection at a facility identified by its FDA Establishment Identifier (FEI) number.

`applications.csv` — Drug application records modeled after FDA's Drugs@FDA database, covering approved and pending applications for generic (ANDA) and brand-name (NDA) drugs, plus a small number of biologics. Each application lists the active ingredient, sponsor, marketing status, and manufacturing facility.

The three files join together: shortage records reference application numbers, and applications reference manufacturing facility FEI numbers, which in turn link to inspection outcomes.

Your Task

Design and build an artifact that allows Dr. Whitmore's staff to ask natural-language questions about drug shortages — and receive answers they can stake their credibility on.

At minimum, your system should:

- **Answer the three required queries below** from the provided data files
- **Show traceability** — specific row-level citations, not just a number
- **Handle ambiguity honestly** — where the data has multiple interpretations, say so and explain the difference rather than picking silently
- **Demonstrate an agentic or workflow-oriented approach** — retrieval, validation, reasoning, or structured intermediate steps

The implementation format is open: a chat UI, CLI, notebook, agent demo, architecture diagram, or lightweight web app. Rough is fine. Something tangible is required.

Required Queries

Your artifact must be able to answer all three of the following. Bring evidence to the walkthrough that it does so correctly.

Query 1:

> *"How many drugs are currently in shortage because of quality problems, and which of their manufacturing sites received an OAI classification in the last two fiscal years?"*

Query 2:

> *"For amoxicillin oral suspension, what is the real supply picture — how many approved applications, how many are actively marketed, and what is currently in shortage?"*

Query 3:

> *"Which foreign manufacturing establishments supplying drugs currently in shortage have not been inspected since FY2021?"*

Note: FDA's fiscal year runs October 1 through September 30. FY2024 = October 1, 2023 through September 30, 2024.

Your tool should be built to handle questions like these — not hardcoded to return these three specific answers. During the walkthrough you may be asked additional queries beyond this list.

Deliverables

For your interview, bring:

1. **A tangible artifact** — prototype application, notebook, architecture diagram, demo workflow, lightweight UI, or CLI. Rough is fine. Something tangible is not optional.
2. **A 5–10 minute walkthrough** — what you built, architectural decisions, where AI and agents were used, where they were intentionally not used, what shortcuts you took, and what you would

improve with another day or week.

3. **A short written or verbal discussion** covering:
4. *Trust and hallucination* — how does your approach reduce the risk of incorrect answers, and where can it still fail?
5. *Agentic workflow design* — why did you choose your structure?
6. *Product thinking* — what type of user is this best suited for today, and what would need to change before it could be used in a real congressional briefing context?

What We Are Evaluating

Not grading on polish. Here is what we are actually looking at:

- **Applied reasoning** — Can you design a practical system around an ambiguous real-world problem?
- **Engineering judgment** — Did you make sensible tradeoffs under time constraints? Can you explain them?
- **Agentic thinking** — Can you structure workflows combining data access, reasoning, and validation coherently?
- **Trust and traceability** — Do you understand why a Senate-testimony context demands a different reliability bar than a consumer chatbot, and how does your approach address it?
- **Communication** — Can you say clearly what your system does, what it does not do, and why?
- **Product intuition** — Did you think about the actual end-user: a policy analyst preparing a briefing under deadline pressure?

Time box: 1–2 hours of implementation time is appropriate. AI-assisted implementation is expected and encouraged. How you use AI tooling is part of what we are evaluating.
